

Basic Concepts in Physics

Force

Definition: Force is a push or pull upon an object resulting from the object's interaction with another object.

Example: Pushing a stalled car to move it forward.

Explanation: When you apply a force to a car, it causes the car to accelerate. The greater the force, the greater the acceleration, following Newton's Second Law ($F = ma$).

Torque

Definition: Torque is a measure of the rotational force on an object.

Example: Using a wrench to loosen a bolt.

Explanation: When you apply force at a distance from the axis of rotation, it creates torque. The greater the force or the longer the handle, the greater the torque ($\text{Torque} = \text{Force} \times \text{Distance}$).

Inertia

Definition: Inertia is the resistance of any physical object to a change in its state of motion or rest.

Example: A book resting on a table remains at rest until someone pushes it.

Explanation: Objects resist changes to their motion. This property is called inertia and is directly related to the mass of the object.

Mass

Definition: Mass is the measure of the amount of matter in an object.

Example: A bag of sugar with a mass of 1 kilogram.

Explanation: Mass is a scalar quantity and does not change regardless of location. It is different from weight, which is the force due to gravity acting on mass.

Work

Definition: Work is done when a force is applied to an object and the object moves in the direction of the applied force.

Example: Lifting a box from the ground to a table.

Explanation: Work is calculated as the product of force and displacement in the direction of the force ($\text{Work} = \text{Force} \times \text{Displacement}$). If there is no movement, no work is done.

Energy

Definition: Energy is the capacity to do work.

Example: A stretched bow has potential energy which can be used to shoot an arrow.

Explanation: Energy comes in many forms such as kinetic, potential, thermal, etc. It can be transformed from one form to another but cannot be created or destroyed (Law of Conservation of Energy).